Technical Supplement: "It's the Elicitation, Not the Eyes: Diagnosing VLM Failures in Slide Quality Inspection and Routing a Symbolic–Neural Critic"

Anonymous Authors (AAAI-27 Submission)

This supplement extends the main paper with material referenced from the body but omitted there for space: the SlideAudit crosswalk (§1), the full multi-reward / VLM-judge audit grid with per-class preference accuracies (§2), the full page-semantic examiner table including the deck-scope negative result (§3), and the extended figures (§4). All numbers are transcribed verbatim from the frozen analysis reports and rollout CSVs that accompany this submission as the Code and Data Supplement; no new claim is made here.

1 SlideAudit crosswalk

Table 1 gives the bidirectional map between our G/S defect classes and the 19-dimensional expert-derived SlideAudit taxonomy, referenced from §3 of the main paper. Empty entries are intentional off-taxonomy semantic or deck-level defects; G7 refines, rather than duplicates, SlideAudit's content-overflow/cut-off category by adding the declared-legal-box criterion that makes the defect linter-blind.

Table 1: Crosswalk between our G/S classes and the 19-dimensional SlideAudit taxonomy.

Our class	SlideAudit class	Relation
G1 text overflow	Content Overflow/Cut-off	equivalent
G2 element overlap	Occluded Content	equivalent
G3 alignment offset	Content Alignment Issues	equivalent
G4 font-size inconsistency	Improper Font Sizing	equivalent
G5 brand-color violation	Excessive or Inconsistent Color Usage; Inappropriate or Mismatched Color Combinations	equivalent
G6 margin violation	Unbalanced Space Distribution	near-equivalent; safe-margin bleed is a special case
G7 render-containment overflow	Content Overflow/Cut-off; related to Improper Image Sizing	our extension; same visual symptom plus declared-legal-box criterion
S1 title/body mismatch	–	off-taxonomy content semantics
S2 narrative-order break	–	off-taxonomy deck-level ordering
S3 terminology inconsistency	–	off-taxonomy cross-slide terminology
S4 density violation	Excessive Text Volume	equivalent
S5 missing logic section	–	off-taxonomy deck-level completeness
S6 image/text contradiction	Irrelevant Visual Content (nearest)	nearest only; contradiction rather than relevance

2 Full multi-reward / VLM-judge audit grid

Table 2 expands the main paper's Table 3 (which reports only G5 and G7) to the full per-class paired preference accuracy $P(r_{\text{clean}} > r_{\text{def}})$ for all audited classes, with 95% CIs on G7. Each scorer penalizes several in-taxonomy defects it can see, confirming each is a functioning scorer rather than a domain mismatch; the G7 column is the linter-blind render class whose detection splits the scorers by perceptual capability, not training domain (§5.3 of the main paper). Per-cell precision and the full multiplicity report (61 paired tests with Holm/Benjamini–Hochberg correction) are in the Code and Data Supplement (`reports/`).

Table 2: Multi-reward / VLM-judge audit: paired preference accuracy per class (chance = 0.5; G7 with 95% CI). ✓/× marks reliable G7 detection (CI above / spanning 0.5). G5 is the internal-chromatic re-operationalisation of §4, a *visible* defect most scorers catch, not the linter-only blind spot G7 is.

Reward scorer	category	G1	G2	G5	**G7**	S4	S6
DocReward-3B	document	0.93	1.00	0.72	× **0.48** [.38, .58]	1.00	1.00
Skywork-VL-7B	general-mm	0.94	0.72	0.57	✓ **0.79** [.69, .86]	0.92	1.00
PickScore-v1	general-mm	0.74	0.59	0.70	✓ **0.71** [.61, .80]	1.00	0.72
LAION-Aesth.	aesthetic	0.37	0.91	0.78	× **0.57** [.46, .66]	0.75	0.61
CLIP-IQA	aesthetic	0.44	0.50	0.65	✓ **0.83** [.74, .90]	0.50	0.72
Qwen3.7-Max judge	frontier VLM	0.13	0.92	–	✓ **1.00** [.86, 1]	1.00	–
n (paired; non-VLM rows)		54	54	80	90	36	18
n (Qwen3.7-Max judge row)		24	24	–	24	24	–

3 Full page-semantic examiner table

Table 3 reproduces the page-semantic inspection table of §6 of the main paper, here shown with the deck-scope S2/S5 negative result and the full per-cell context.

Table 3: Page-semantic inspection, balanced accuracy [95% Wilson] on paired clean (best modality). The finetuned 8B examiner beats both zero-shot baselines, including the 30B, *in-distribution*. Deck-scope S2/S5 are a negative result. *n* = paired positives (2-AFC rows = forced-choice pairs); the S6 2-AFC rests on only 12 pairs and is descriptive only.

class	**ft-8B**	zs-8B	zs-30B	*n*
S1 title/body	**1.000** [.95,1]	0.778 [.69,.85]	0.933 [.87,.95]	36
S4 density	**1.000** [.97,1]	0.529 [.50,.58]	0.774 [.69,.84]	60
G1 overflow (2-AFC)	**1.000** [.97,1]	0.975 [.93,.99]	0.700 [.61,.77]	120
S6 image/text (2-AFC)	**1.000** [.76,1]	1.000 [.76,1]	0.500 [.25,.75]	12
S2 narrative (deck)	0.500† [.45,.53]	0.646 [.53,.72]	0.549 [.42,.66]	36
S5 missing-logic (deck)	0.500 [.47,.53]	0.500 [.47,.53]	0.567 [.45,.68]	60

† degenerate (always-flag): the SFT mix had no clean-*deck* negatives. See §6 of the main paper.

4 Extended figures

These figures were omitted from the main paper for space; their findings are stated in the body prose. They are included here for completeness.

5 Conceptual figures

These vector “concept” figures (with text overlays) were used in the single-column technical report and are reproduced here for readers who prefer the pictorial version of the protocol and routing arguments. They

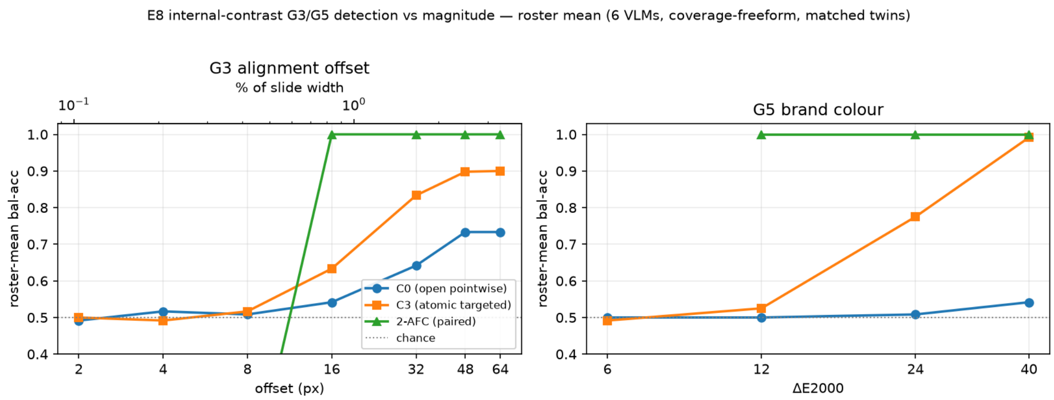

Figure 1: Fine geometry is magnitude-gated, not a flat capability floor (referenced as the saturation curve in §4). Roster-mean balanced accuracy (6 VLMs, matched-twin internal-contrast set) vs. injected magnitude for G3 alignment (left) and G5 brand color (right). Open pointwise (C0) lags; atomic targeted (C3) recovers to ≈ 0.90 by 48–64 px / 0.99 by ΔE 40; the paired 2-AFC saturates to 1.00 by 16 px (0.83%) / ΔE 12. The sub-threshold tail (≤ 8 px, $\leq 6\,\Delta E$) stays at chance under every protocol.

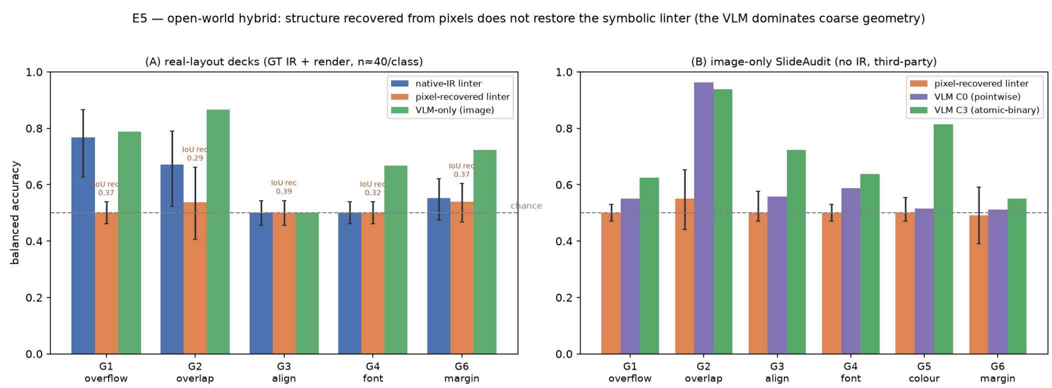

Figure 2: Open-world hybrid: structure recovered from pixels does not restore the symbolic linter (referenced in §7). **(A)** real-layout decks (GT IR + render, $n \approx 40$/class): native-IR linter vs. pixel-recovered linter vs. VLM-only floor, Wilson CIs; recovered structure rescues only overlap (G2) and is silent on fine geometry, while the VLM dominates every class. **(B)** image-only SlideAudit (no IR, third-party): the recovered linter clears chance only on G2; the C3-elicited VLM is far stronger.

convey no information beyond the body prose and the data figures above; they are included for readability.

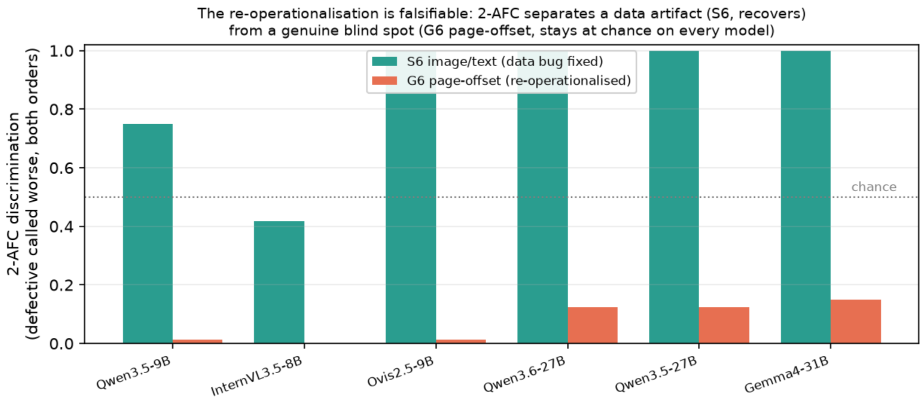

Figure 3: A genuine blind spot the forced choice exposes (referenced in §4). Per-model 2-AFC discrimination. Pointwise puts both classes at chance, but the forced choice *separates* them: the data artifact S6 recovers on every model, while G6 page offset stays at the floor on every model—a genuine blind spot, not an elicitation or data artifact.

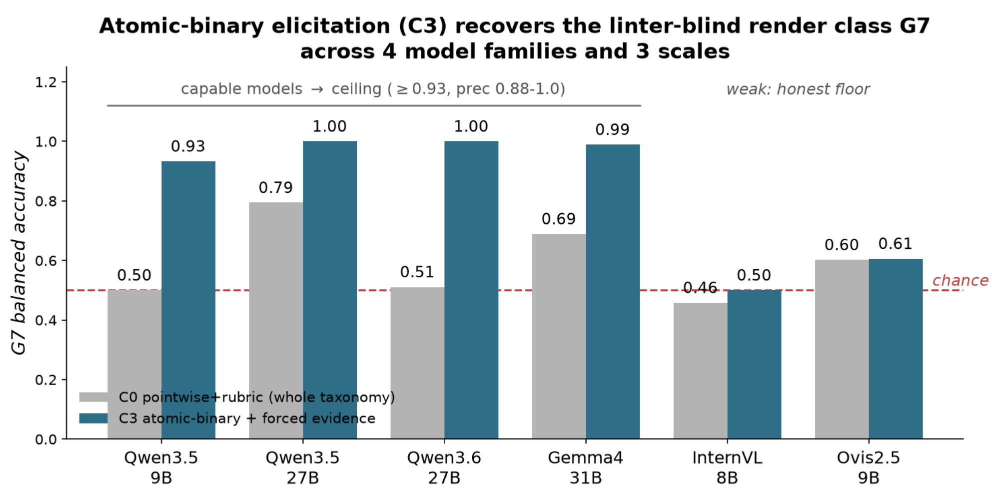

Figure 4: Atomic-binary elicitation (C3) recovers the linter-blind render class G7 across the capable subset (referenced in §5.1). Capable models reach the ceiling (≥ 0.93, precision 0.88–1.0); the weaker InternVL/Ovis stay near their perceptual floor, a capability-dependent boundary, not a universal fix.

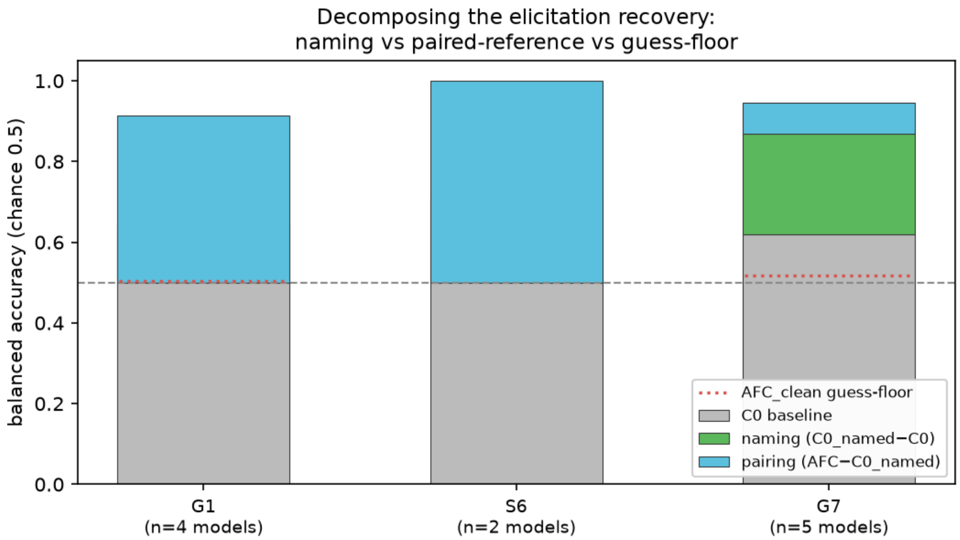

Figure 5: What drives the recovery: naming vs. a paired reference (referenced in §5.1). The chance→ceiling jump factors into a *naming* component (C0_named−C0) and a *pairing* component (AFC−C0_named). For G7 the naming term carries it (green); for G1 and S6 the paired reference does (blue). The AFC_clean guess-floor (dotted) is near chance, so neither is a forced-pick artifact.

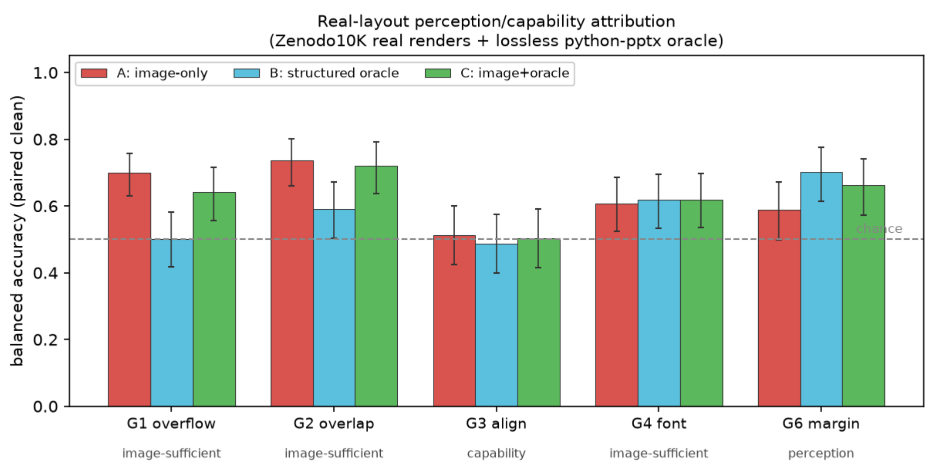

Figure 6: The perception/capability split reproduces on real layouts (referenced in §7). ZENODO10K real renders + lossless `python-pptx` oracle; mean over 3 VLMs / 3 families, balanced accuracy on paired clean; 95% Wilson on counts pooled across models (n=76–90 per class). Coarse defects are image-sufficient; G6 margin is a perception bottleneck the oracle rescues (B>A); G3 alignment stays near chance in every channel on real cluttered decks.

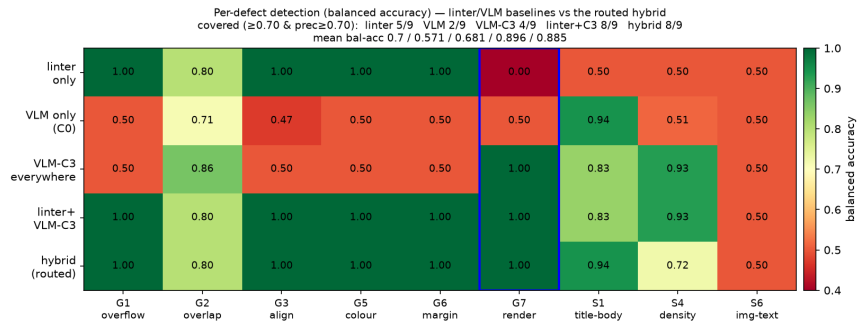

Figure 7: Coverage by engine (companion to Table 2 of the main paper). Per-defect balanced accuracy for the linter, VLM-C0, VLM-C3, linter+VLM-C3, and the routed hybrid. Minimal linter⊕C3 matches routed coverage; G7 (boxed) is the linter-blind cell rescued by C3.

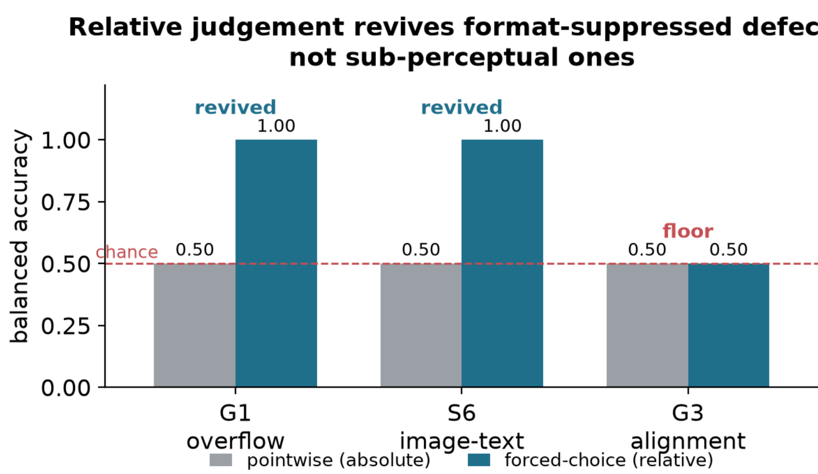

Figure 8: Relative judgement recovers absolute-pointwise blind spots (referenced in §4). For G1 overflow and S6 image/text contradiction, pointwise (absolute) inspection is at chance but a forced choice recovers perfect detection. G3 alignment behaves the same way above a magnitude threshold; only its sub-threshold tail and S3 terminology stay at chance side by side.

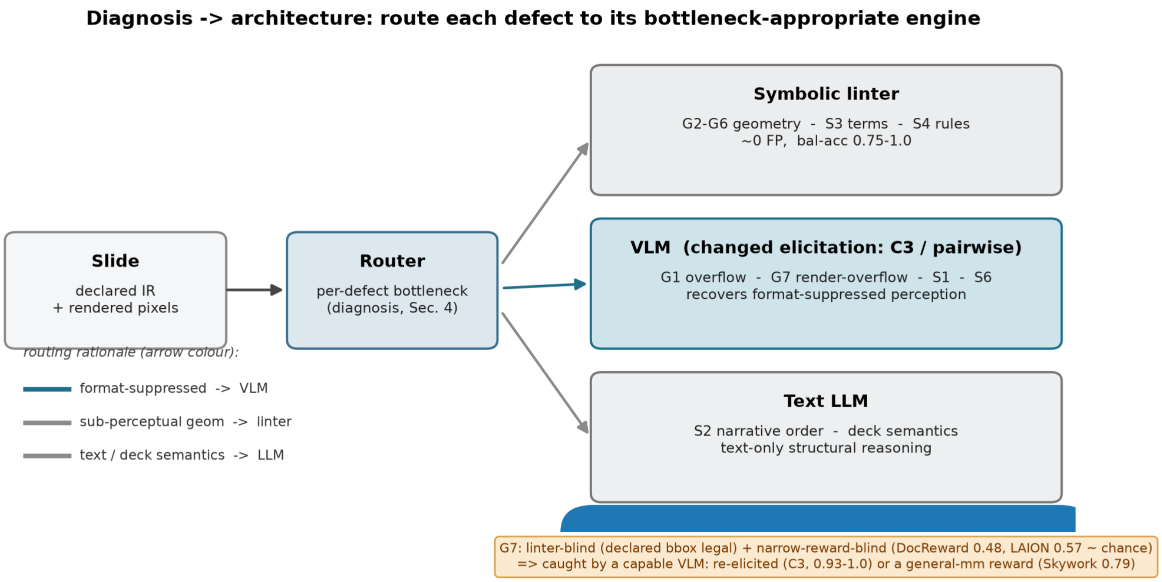

Figure 9: Diagnosis → architecture. Route each defect to the engine that owns its bottleneck: declared geometry to a symbolic linter, format-suppressed perceivable classes to a re-elicited VLM, text/deck semantics to an LLM. The G7 callout marks the linter-blind render-overflow class caught only by an engine with a rendered-quality read-out (Sec. 5 of the main paper). Companion to the teaser (Fig. 1).

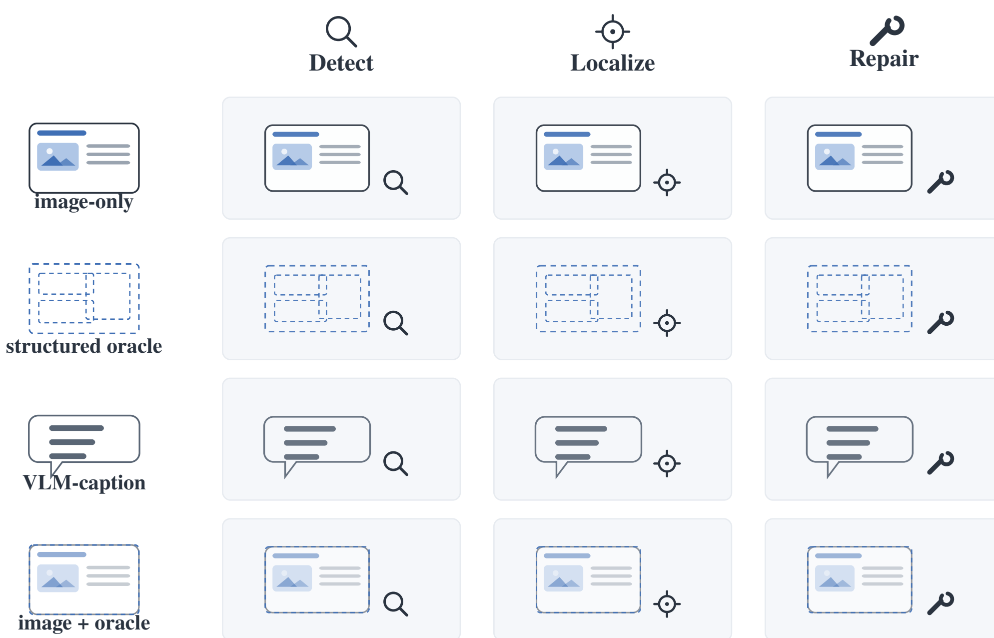

Figure 10: Attribution protocol. Each paired (defective, clean) slide is presented under four input modalities (image-only, the lossless structured oracle, a fixed VLM-caption control, and image+oracle) crossed with detect/localize/repair. Comparing image-only against the structured oracle isolates a clean perception/reasoning split (Sec. 3 of the main paper).

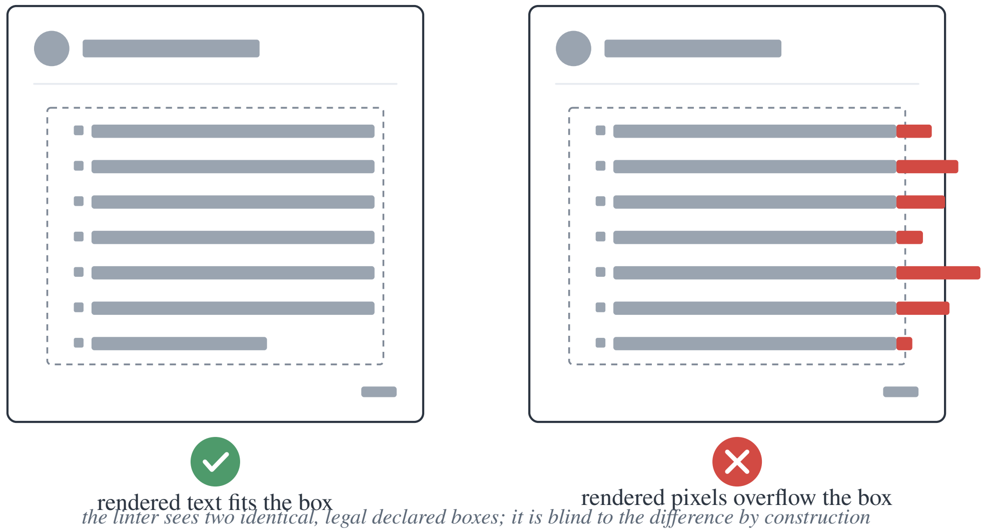

Figure 11: G7 render-overflow. The declared box (dashed) is legal and *identical* in both slides; only the rendered pixels overflow on the right. The linter and any structure-only model read the same legal box in both, so they are blind to G7 by construction (Sec. 5.3 of the main paper).

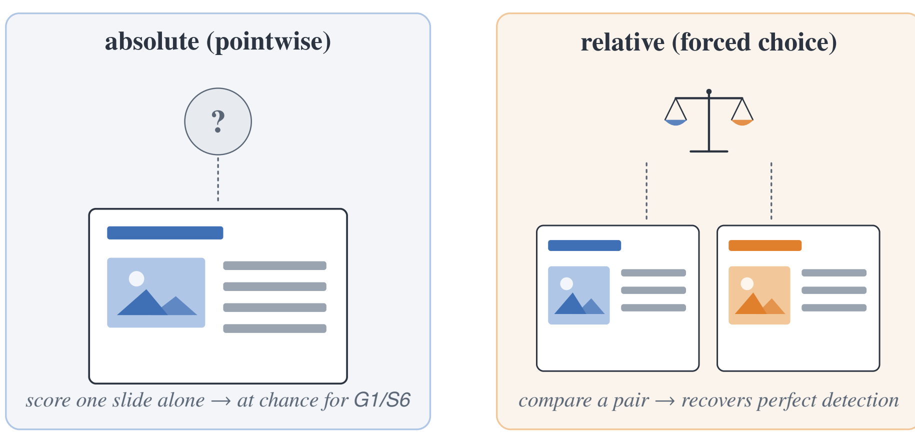

Figure 12: Two elicitations, same model and images. Pointwise (absolute) inspects one slide in isolation; the forced choice compares a pair. For G1 and S6, changing absolute to relative is what recovers detection (Sec. 4 of the main paper).

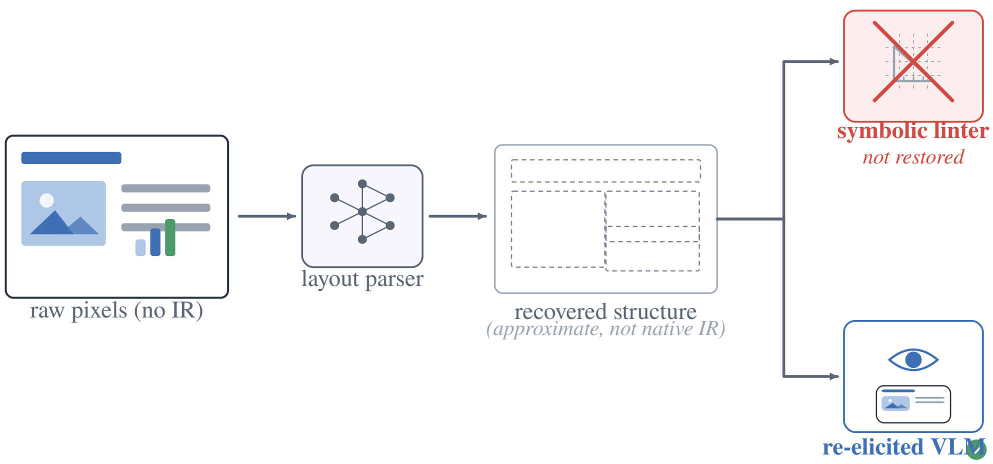

Figure 13: Open world: recovered structure does not restore the linter. For bare third-party pixels with no IR, a layout parser recovers approximate structure, but it is not the native IR the symbolic linter needs; only the re-elicited VLM remains a deployable critic (Sec. 7 of the main paper).